Ch 15 - Constrained Optimization

Case 1: objective function and constraints are linear.
(Linear Programming)

Case 2: Nonlinear constrained optimization.

Linear Programming

objective function:

$$\text{maximize} \quad z = c_1 x_1 + c_2 x_2 + \dots + c_n x_n$$

(profit)

c_j: payoff of each unit of j'the activity.

x_j: magnitude of jth activity.

z: total payoff due to n activities.

constraints: $a_{i1} x_1 + a_{i2} x_2 + \dots + a_{in} x_n \leq b_i$

b_i: amount of the ith resource that is available (Resources are limited)

a_{ij}: amount of the ith resource that is consumed for each unit of jth activity.

constraint: $x_i \geq 0$

Ex Gas-processing plant: regular (grade) quality or premium quality

	Product		
Resource	Regular	Premium	Resource Availability
Raw Gas	7 m^3/tonne	11 m^3/tonne	77 m^3/week
Production Time	10 hr/tonne	8 hr/ton	80 hr/week
Storage	9 ton	6 ton	
Profit	150/ton	175/ton	

- Only one of the grades can be produced at a time
- Both grades are guaranteed to sell.

Develop a LP formulation to maximize the profit for this operation.

Soln How much of each gas should be produced to maximize the profit?

Total profit $= 150x_1 + 175x_2 = z$

$$\boxed{\text{maximize } z = 150x_1 + 175x_2}$$ objective function

Total gas used = constraints

$7x_1 + 11x_2 \leq 77$ (1) constraint (material)

$10x_1 + 8x_2 \leq 80$ (2) time constraint

$x_1 \leq 9$ (3) regular storage constraint

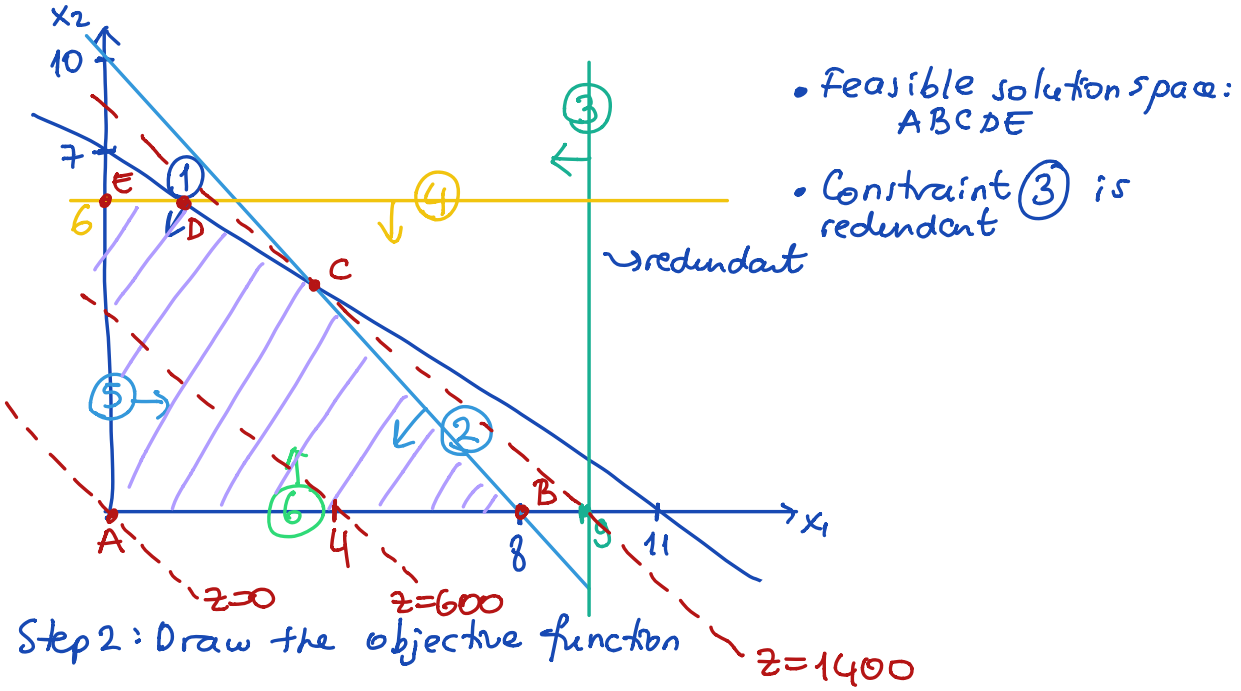
$x_2 \leq 6$ (4) premium " "

$x_1 \geq 0$ (5) } positivity constraints

$x_2 \geq 0$ (6) }

Graphical Solution

Step 1 : Draw the constraints



- Feasible solution space: ABCDE
- Constraint (3) is redundant

Step 2 : Draw the objective function

At $z=0 = 150x_1 + 175x_2 \Rightarrow x_2 = \frac{-150}{175}x_1$

at $z=600 = 150x_1 + 175x_2$

$$x_2 = \frac{600}{175} - \frac{150}{175}x_1$$

at $z=1400 = 150x_1 + 175x_2 \Rightarrow \left.\begin{array}{l} x_1 = 4.9 \\ x_2 = 3.9 \end{array}\right\}$ max. profit is 1400

Substitute the answer into constraints:

$\left.\begin{array}{l} 7(4.9) + 11(3.9) \cong 77 \\ 10(4.9) + 8(3.9) \cong 80 \end{array}\right\}$ just meet the resource and time constraints. (binding constraints)

$\left.\begin{array}{l} 4.9 \leq 9 \\ 3.9 \leq 6 \end{array}\right\}$ non-binding

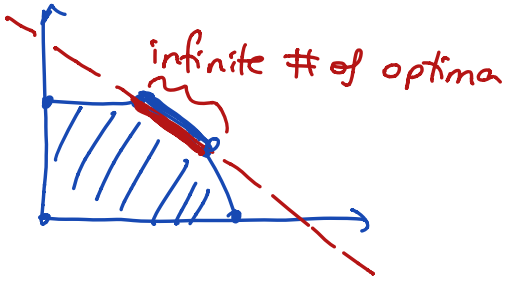
Conclusions

- can increase profit by by increasing raw gas and production time resources.
- increasing storage will have no impact on profit.

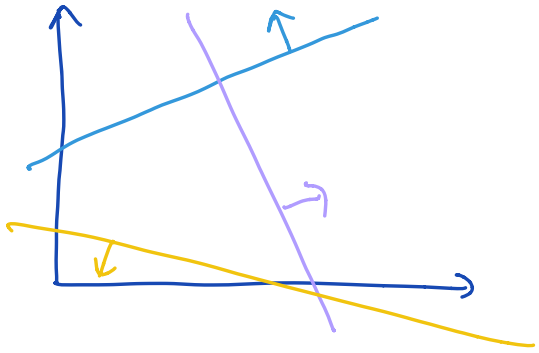
* Four possible outcome in a linear programming problem

1) unique solution: a single pt (prev. example)

2) alternate solutions: objective function is parallel to one of the constraints.



3) No feasible solution - problem is over constrained



4) Unbounded Problems : problem is underconstrained

